AR-Mini · T2 · Safety layer

firmware/armini/lib/core/Safety.h / .cpp · decides every tick whether the robot may run · numbers = SafetyConfig defaults (VERIFY) · grey mono = dry-run command · v1 · 10.10.2026

1 · Four causes, each a small state machine

KILL

kill_raw: rocker in KILL (GPIO14 low) · or ladder < 0.6 V

`kill 1 / kill 0`

off — rocker in RUN

rocker → KILL · no debounce

ACTIVE — the same tick

rocker → RUN

LEAK

leak_raw: probe wet (GPIO7) · or ladder 0.6–2.0 V

`leak 1 / leak 0`

dry

probe wet

wet · timing — must stay wet

100 ms wet — `a splash does not trip`

ACTIVE

dry

BATTERY

v_bat = v_adc × 133 / 33 (divider 100 k / 33 k)

`vbat 9 / vbat 11`

OK

v_bat < 9.9 V

dip · timing — v_bat < 9.9 V

500 ms low — `one bad sample does not trip`

LOW = ACTIVE — stays LOW between 9.9 and 10.5 V

v_bat > 10.5 V

HEARTBEAT

heartbeat_ok: the main loop / link reported in time

`hb 0 / hb 1`

ok

heartbeat missing

missing · timing

500 ms missing

LOST = ACTIVE

heartbeat ok

priority

If several are active in the same tick, the highest wins:

1. KILL
2. LEAK
3. BATTERY
4. HEARTBEAT

→ active = that cause, or NONE

active

2 · The verdict (a latch)

OK

- verdict.safe = true
- The robot may run.

active ≠ NONE → trip — `kill 1 · leak 1 · vbat 9 · hb 0`

reset() and active = NONE — `kill 0 → reset`

TRIPPED

- verdict.safe = false
- reason = the FIRST cause, kept until reset
- A cause that clears by itself does NOT restart the robot.

reset() while a cause is active → refused, stays TRIPPED

→ T1 mission state machine

applyVerdict(): exactly one of kill / leak / battery_low / heartbeat_lost is true while TRIPPED. T1 jumps to SAFE(reason) in the same tick. The raw sensors never reach T1.

3 · Where the inputs come from

DevKit-Lipo (now)	leak on GPIO7, kill on GPIO14, VBAT on GPIO4 — separate pins, no ladder (ladder_v = −1).
XIAO carrier (later)	leak + kill share one ADC pin A1 through a resistor ladder: 3.30 V OK · 1.06 V LEAK · 0.30 V KILL · 0.25 V both → KILL. Thresholds < 0.6 V KILL, 0.6–2.0 V LEAK, > 2.0 V OK.
First line of defence	hardware: the rocker in KILL cuts the motor drivers without the MCU (BTS7960 R_EN/L_EN, PCA9685 OE). T2 makes the firmware agree and adds what hardware cannot see.
Fail-safe	v_bat = 0 V (unknown) trips BATTERY after 500 ms. A missing heartbeat trips HEARTBEAT.